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Marks 1	3 questions
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Answer key	at the end

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Marks 1 3 questions

Q1. GATE ECE 2021

The refractive indices of the core and cladding of an optical fiber are 1.50 and 1.48 respectively. The critical propagation angle. Which is defined as the maximum angle that the light beam makes with the axis of the optical fiber to achieve the total internal reflection, (rounded off to two decimal places) is ____ degree.

Open solution

Q2. GATE ECE 2017 Set 1

Consider a wireless communication link between a transmitter and a receiver located in free space, with finite and strictly positive capacity. If the effective areas of the transmitter and the receiver antennas, and the distance between them are all doubled, and everything else remains unchanged, the maximum capacity of the wireless link

Open solution

(A) increases by a factor of 2

(B) decrease by a factor 2

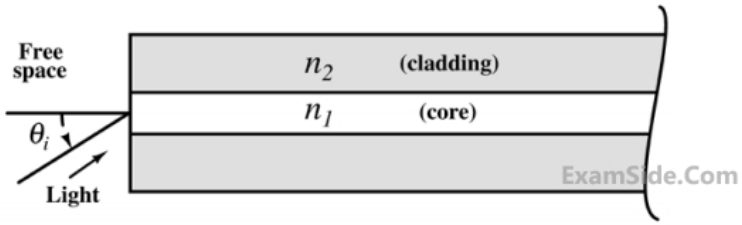
(C) remains unchanged

(D) decreases by a factor of $\sqrt{2}$

Q3. GATE ECE 2016 Set 2

Light from free space is incident at an angle θ_1 to the normal of the facet of a step-index large core optical fibre. The core and cladding refractive indices are $n_1 = 1.5$ and $n_2 = 1.4$ respectively.

Open solution



The maximum value of θ_1 (in degrees) for which the incident light will be guided in the core of the fibre is _____ .

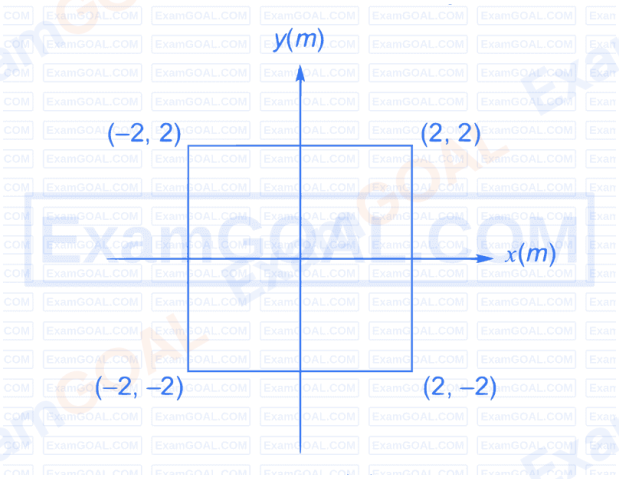
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Marks 2 4 questions

Q4. GATE ECE 2025

Open solution

A square metal sheet of $4\,\mathrm{m} \times 4\,\mathrm{m}$ is placed on the $x-y$ plane as shown in the figure below. If the surface charge density (in $\mu\mathrm{C/m^2}$) on the sheet is $\rho_s(x, y) = 4|y|$, then the total charge (in $\mu\mathrm{C}$, rounded off to the nearest integer) on the sheet is_____ .

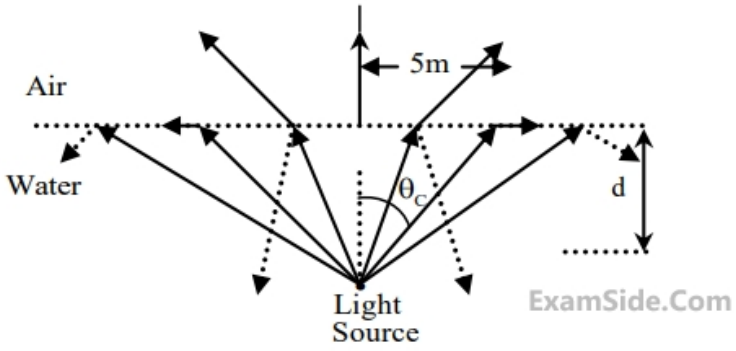


(A) 16 (B) 85 (C) 64 (D) 256

Q5. GATE ECE 2017 Set 2

Open solution

The permittivity of water at optical frequencies is $1.75\varepsilon_0$. It is found that an isotropic light source at a distance d under water forms an illuminated circular area of radius $5m$, as shown in the figure. The critical angle is θ_c



The value of d (in meter) is _____ .

Q6. GATE ECE 2017 Set 1

Open solution

An optical fiber is kept along the $\widehat{Z}$ direction. The refractive indices for the electric fields along $\widehat{X}$ and $\widehat{Y}$ directions in the fiber are $n_x = 1.5000$ and $n_y = 1.5001$, respectively ($n_x \neq n_y$ due to the imperfection in the fiber cross-section). The free space wavelength of a light wave propagating in the fiber is $1.5\ \mu m$. If the light wave is circularly polarized at the input of the fiber, the minimum propagation distance after which it becomes linearly polarized, in centimeters, is _____ .

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Q7. GATE ECE 2016 Set 3

A radar operating at 5 GHz uses a common antenna for transmission and reception. The antenna has a gain of 150 and is aligned for maximum directional radiation and reception to a target 1 km away having radar cross-section of $3m^2$. If it transmits 100 kW, then the received power (in μW) is _____ .

Open solution

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Answer Key

Marks 1

1. **9.36**	2. **C**	3. **32.5**

Marks 2

4. **C**	5. **4.33**	6. **0.375**	7. **0.0122**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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